

Lab 03 — First GoogleTest Project: Calculator

1. Objectives

By the end of this lab you will be able to:

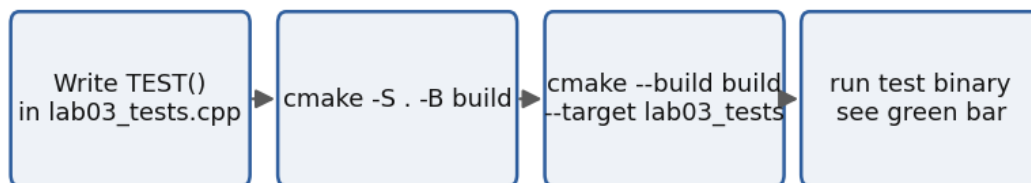
1. Configure a small CMake project that builds and runs a GoogleTest binary.
2. Write `TEST()` blocks with `EXPECT_EQ` assertions.
3. Read the coloured summary line and interpret pass / fail counts.

The concept focus this week is the **toolchain** — no coverage theory beyond what you already saw in Lab 2. Prime paths were discussed in the lecture 2 and will be revisited in Assignment 1.

2. Learning outcomes

- **LO1** — Run `cmake -S . -B build` and `cmake --build build` in sequence.
- **LO2** — Write a passing GoogleTest case for a simple function.
- **LO3** — Interpret the GoogleTest output.

3. Workflow



CMake fetches GoogleTest automatically on first configure.

4. Problem statement

You are given a tiny Calculator class with two methods:

- `int add(int a, int b)`
- `int subtract(int a, int b)`

There is one worked test in `test/lab03_tests.cpp` (`Add_ReturnsSumForPositive`) so the first build shows a green bar immediately. Your job is to add tests for the missing cases and see the summary line grow.

5. Tasks

1. Copy the starter into `studentID/LabNN/lab03/`.

2. Configure and build (first run downloads GoogleTest — this can take one to two minutes):

```
cmake -S . -B build
cmake --build build --target lab03_tests
```

3. Run the binary and confirm the one included test passes:

```
./build/lab03_tests --gtest_color=yes
```

4. Complete the four TODO tests in test/lab03_tests.cpp:

- Add_ReturnsSumForNegatives
- Subtract_ReturnsDifference
- Subtract_ReturnsZeroForEqualInputs
- Add_HandlesZero

5. Rebuild and rerun. Aim for a green bar with 5 passing tests.

6. Standard submission

1. **Updated source code** — src/calculator.{hpp,cpp}, test/lab03_tests.cpp, CMakeLists.txt.
2. **VSCode screenshot** — screenshots/vscode.png.
3. **Terminal screenshot** — screenshots/terminal.png (must show the coloured summary line).
4. **Lab report** — Lab03_Report.md, .docx, or .pdf. 150–250 words.

Your lab report must detail the steps you took to install, configure, and test your environment. Ideally, your report should include the screenshots of your workflow. Finally, conclude your assessment how using a framework could help you scale up your tests.

Submission Deadline: Tuesday, 29 September 2026 (1700 hrs)